

inbuilt comparator, internal timer and external interrupt, have been enabled.

In the main part of the software, the MCU scans the status of DIP1 and stores the corresponding startup delay. On pressing S3, the software activates both solid-state relays to connect the power supply to the motor through start-and-run capacitors.

The software monitors the current output from CT1. If current is less than the set value, it assumes that the motor is delivering very low current. Connections will not be successful if current is low. The software deactivates all outputs (pins 16 and 17 of IC2). The motor pump stops working, and motor failure indicator gets turned on.

If current output is found normal, the system switches off the start winding output after a set delay. The motor pump keeps running, and motor current output is monitored from CT1.

If motor current is found more than the overload set value and the same condition stays for a predetermined time, overload condition gets

Semiconductors:

IC1	- 7805, 5V voltage regulator
IC2	- AT89C4051 MCU
IC3-IC4	- MOC3041 optocoupler
T1	- BC548 npn transistor
T2, T3	- BC557 pnp transistor

TRIAC1,

TRIAC2	- BT139/BTA41 triac
BR1	- W10 bridge rectifier
D1, D2	- 1N4007 rectifier diode
ZD1	- 2.1V Zener diode
ZD2	- 5.1V Zener diode
LED1-LED5	- 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1, R5	- 10-kilo-ohm
R2, R4, R7, R8,	
R10	- 2.2-kilo-ohm
R3	- 8.2-kilo-ohm
R6, R9, R11	- 5.6-kilo-ohm
R12	- 680-kilo-ohm
R13, R17	- 330-ohm
R14, R15	- 220-ohm
R16, R19	- 390-ohm
R18, R20	- 39-ohm
VR1, VR2	- 100-kilo-ohm potmeter
VR3	- 10-kilo-ohm potmeter

Capacitors:

C1	- 470 μ F, 25V electrolytic
----	---------------------------------

PARTS LIST

C2, C4	- 10nF ceramic disk
C3	- 100 μ F, 16V electrolytic
C5	- 10 μ F, 16V electrolytic
C6	- 100 μ F, 25V electrolytic
C7, C8	- 22pF ceramic disk
Miscellaneous:	
CON1, CON2,	
CON4, CON5	- 2-pin connector
CON3	- 3-pin connector
CON6	- 2-pin terminal connector
F1	- 10A fuse with holder
XTAL1	- 12MHz crystal
S1	- On/off switch
S2-S4	- Tactile switch
DIP1	- 3-way dip switch
V	- Voltmeter
A	- Ammeter
B1, B2	- 100W, 230V AC bulb for testing
CT1	- 10A, 50Hz current transformer
X1	- 230V AC primary to 9V-0-9V, 500mA secondary transformer
	- 1/1.5HP motor with start-and-run capacitor
	- 230V AC mains power supply

activated and all outputs are deactivated. The motor pump stops.

Internal timer function is used to

get accurate long delay, and interrupt function is used for switching off the motor at any time without any delay.

support@efy.in

Do you have a query, suggestion or a complaint?

You can e-mail it to
support@efy.in
and we will take care
of the rest.



electronics
FOR YOU
www.electronicsforu.com

COMPONENTS PRODUCTS MACHINERY
ELECTRONICS
BAZAAR
www.eb.efyindia.com

OpenSource
THE COMPLETE MAGAZINE ON OPEN SOURCE
For You
www.opensourceforu.com

EFYGROUP
Technology Drives Us
www.efy.in